

BESANT TECHNOLOGIES



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Technologies**

Title:

Fitness Data Analysis for Health and Performance Insights

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1. Introduction

The project focuses on analyzing a fitness dataset to understand the relationship between physical attributes, dietary habits, and workout performance. By studying key indicators such as age, BMI, heart rate (BPM), and calories burned, this analysis aims to uncover meaningful health patterns and fitness trends. The analysis leverages **Python**, **Pandas**, and **data visualization libraries** like **Matplotlib** and **Seaborn** to perform a detailed exploration of the dataset, identify trends, and provide actionable insights that can help individuals optimize their workout routines.

2. Objective of the Analysis

The main objective of this analysis is to:

- Understand the fitness patterns of users based on key metrics like Calories Burned, BMI, BPM, and Workout Type.
- Explore how age, gender, and workout frequency affect calorie expenditure and performance.
- Identify correlations between dietary factors (protein, fat, carbs) and physical activity outcomes.
- Generate insights that can guide personalized fitness recommendations.

2.1 Problem Statement

Despite the availability of fitness data, many individuals struggle to interpret it effectively. They often fail to recognize how factors like **diet composition**, **heart rate**, and **exercise intensity** interact to influence overall fitness outcomes.

This project addresses this gap by performing an analytical study on the dataset to answer key questions such as:

- Which workout type leads to the maximum calorie burn?
- How does heart rate (BPM) relate to workout intensity?
- Does higher protein intake contribute to improved fitness performance?
- How do BMI and age affect energy expenditure?

By answering these questions, this project enables data-driven decision-making in fitness planning and health optimization.

2.2 Key Questions / KPIs

Key Question	Description	Purpose / Objective
What is the average age of users participating in fitness activities?	Finds the age range and mean age of the users in the dataset.	To understand the target age group most involved in fitness.
What is the average BMI of users, and how does it relate to calorie burn?	Checks whether BMI affects calories burned.	To study how body composition influences workout results.
Which type of workout is most common among users?	Identifies the most frequently performed exercise types.	To find user preferences and workout trends.
How many calories are burned on	Calculates average calories	To measure workout effectiveness and energy expenditure.

Key Question	Description	Purpose / Objective
average during each workout session?	burned across all users.	
What is the average Max BPM and Resting BPM among users?	Examines heart rate behavior during and after exercise.	To assess users' cardiovascular strength and recovery ability.
Does a higher Max BPM result in more calories burned?	Compares high heart rate sessions with calorie burn data.	To analyze the impact of workout intensity on calories burned.
Is there a relationship between protein intake and calories burned?	Looks at how protein intake levels affect workout results.	To evaluate the role of nutrition in improving fitness performance.
Which age group tends to burn the most calories?	Groups users by age and compares calorie burn.	To identify which age range shows better energy output.
How does Workout Type affect Resting BPM and Average BPM?	Compares heart rate data for different exercise types.	To understand which workouts improve heart health the most.

3. Data Collection

In this project, the dataset used was named “lifestyle_data.csv”, which contains detailed information

related to users' health and fitness attributes such as Age, Gender, BMI (Body Mass Index), Resting BPM (Beats Per Minute), Max BPM, Calories Burned, Workout Type, Workout Frequency, Protein Intake, and other related variables.

3.1 Data Workflow:

1. The raw data was imported into MySQL for structured management and query validation.
2. The cleaned dataset was loaded into Jupyter Notebook using Python's Pandas library.
3. Data preprocessing and transformation steps were performed to prepare it for visualization and analysis.

Before analysis, it is important to understand the structure and quality of the dataset.

Column Name	Description
Age	The age of the individual
Gender	Gender of the user (Male/Female)
Height_cm	Height of the user measured in centimeters.
Weight_kg	Weight of the user measured in kilograms.
BMI	Body Mass Index calculated using weight and height.
Resting_BPM	Heart rate in beats per minute when the person is at rest.
Max_BPM	Maximum heart rate achieved during exercise.

Avg_BPM	Average heart rate maintained during workout.
Calories_Burned	Total calories burned during the workout session.
Workout_Type	The category or type of workout performed (e.g., Cardio, Strength, Yoga).
Workout_Frequency	How often the person works out (e.g., Daily, 3 times/week, Weekly).
Workout_Duration	Duration of the workout session in minutes.
Steps_Taken	Number of steps recorded during the workout session.
Distance_km	Distance covered during the workout in kilometers.
Protein_per_kg	Protein intake per kilogram of body weight (grams per kg).
Water_Intake_L	Amount of water consumed (in liters) during or after the workout.
Sleep_Hours	Total sleep duration of the user in hours.
Workout_Date	Date on which the workout data was recorded.
Goal	The user's fitness goal (e.g., Weight Loss, Strength, Endurance).
Trainer_Assigned	Indicates if a personal trainer is assigned (Yes/No).
Membership_Type	Type of membership (e.g., Basic, Premium, VIP).
Calories_Intake	Total calories consumed per day (approximation).
Fat_Percentage	Percentage of body fat measured.

Carbs_Percentage	Percentage of carbohydrate intake per day.
Protein_Percentage	Percentage of protein intake per day.

3 Purpose of Column Classification

This classification helps determine which analysis and visualization methods to use:

- Numerical variables — analyzed using histograms, scatter plots, and boxplots.
- Categorical variables — analyzed using count plots, bar charts, and pie charts.

Numerical Columns: Age, Height_cm, Weight_kg, BMI, Resting_BPM, Max_BPM, Avg_BPM, Calories_Burned, Workout_Duration, Steps_Taken, Distance_km, Protein_per_kg, Water_Intake_L, Sleep_Hours, Calories_Intake, Fat_Percentage, Carbs_Percentage, Protein_Percentage

Categorical Columns: Name, Gender, Workout_Type, Workout_Frequency, Workout_Date, Goal, Trainer_Assigned, Membership_Type

4. Data Cleaning & Transformation

Data cleaning and transformation are important steps to make sure the dataset is accurate, consistent, and ready for analysis. The main goal is to remove errors, handle missing values, and convert data into proper formats that can be used for visualization and modelling.

Step 1: Check for missing values.

Step 2: Fill or remove missing data.

Step 3: Check and remove duplicate rows.

Step 4: Correct wrong data types (e.g., change date to datetime).

Step 5: Standardize text values (like Gender, Workout Type).

Step 6: Handle outliers (remove or adjust extreme values).

Step 7: Rename columns if needed.

Step 8: Create new useful columns (like BMI category).

Step 9: Save the cleaned dataset as a new file.

5. Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) helps to understand the structure and key patterns of the dataset.

It involves studying the data using statistics and visualizations to find trends, relationships, and possible errors.

Both univariate and bivariate analysis were performed in this project using Matplotlib and Seaborn libraries.

Step 1: Understanding the Data

- Used `d.info()` to view data types and missing values.
- Used `d.describe()` to check numerical summaries such as mean, median, min, and max.
- Checked the number of rows and columns to understand dataset size.

Step 2: Univariate Analysis

Univariate analysis studies one variable at a time.

- For numerical columns (like Age, BMI, Calories_Burned), histograms and boxplots were used to see the data distribution and outliers.

- For categorical columns (like Gender, Workout_Type), count plots and bar charts were used to see which categories are most common.

Step 3: Bivariate Analysis

Bivariate analysis shows relationships between two variables.

- Scatter plots were used to compare continuous values (e.g., Age vs Calories_Burned).
- Boxplots were used to compare categories with numerical data (e.g., Workout_Type vs Resting_BPM).

6. Data Visualization and Insights

Data visualization helps in understanding patterns and relationships within the dataset effectively.

Using Matplotlib and Seaborn, different charts were plotted for both numerical and categorical data.

A total of 10 visualizations were created, and their insights are explained below

7. Insight Generation

After the visualization phase, several meaningful insights were drawn:

1. Age vs Calories Burned:
Younger individuals tend to burn slightly more calories, possibly due to higher metabolism and activity levels.
2. Workout Type:
HIIT and Strength Training show the highest calorie burn, while Yoga and Cardio sessions have moderate results.

3. BMI Distribution:

The majority of users fall under the normal BMI range (18–25), suggesting a relatively healthy population.

4. Heart Rate Insights:

Individuals with higher Max BPM burn more calories, indicating that workout intensity has a direct impact on calorie expenditure.

5. Dietary Correlations:

Users with higher protein intake per kg show better calorie-burning performance, emphasizing the importance of balanced nutrition.

6. Gender Comparison:

Both males and females engage equally in workouts, but the average calorie burn slightly differs due to body composition.

7. Resting BPM vs Workout Type:

Regular exercisers have lower resting BPM, showing better cardiovascular endurance.

8. Correlation Heatmap:

A strong positive correlation was observed between Max BPM and Calories Burned, and a moderate correlation between Protein Intake and Calories Burned.

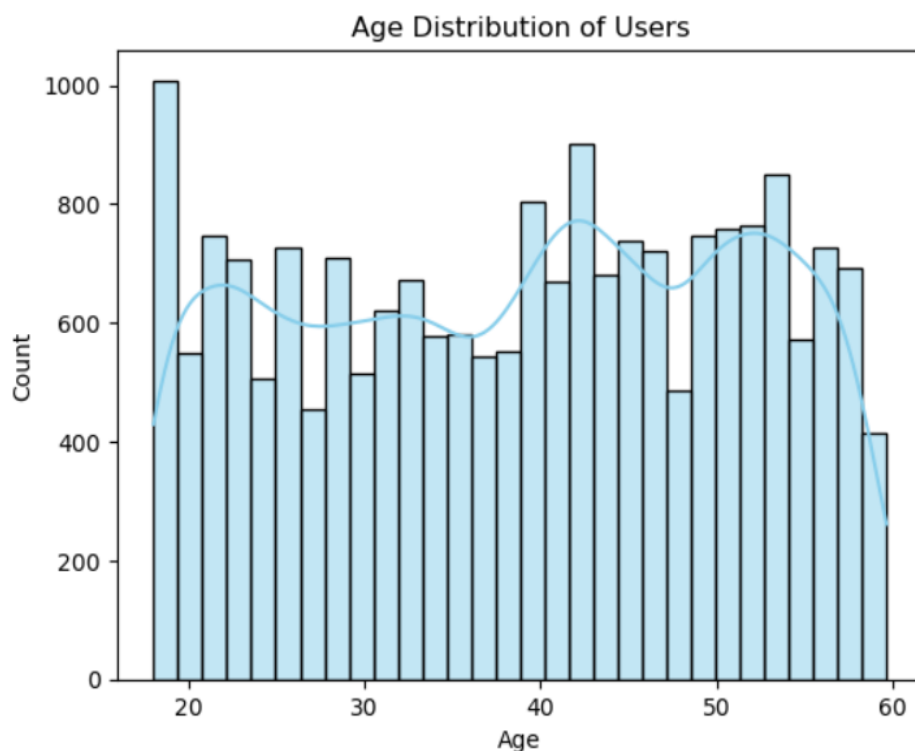
8. Tools and Technologies Used

Tool / Library	Purpose
Python (Pandas, NumPy)	Data handling & preprocessing
Matplotlib & Seaborn	Data visualization

Tool / Library	Purpose
MySQL	Data storage & retrieval
Jupyter Notebook	Data analysis environment
MS Word / PowerPoint	Reporting & presentation

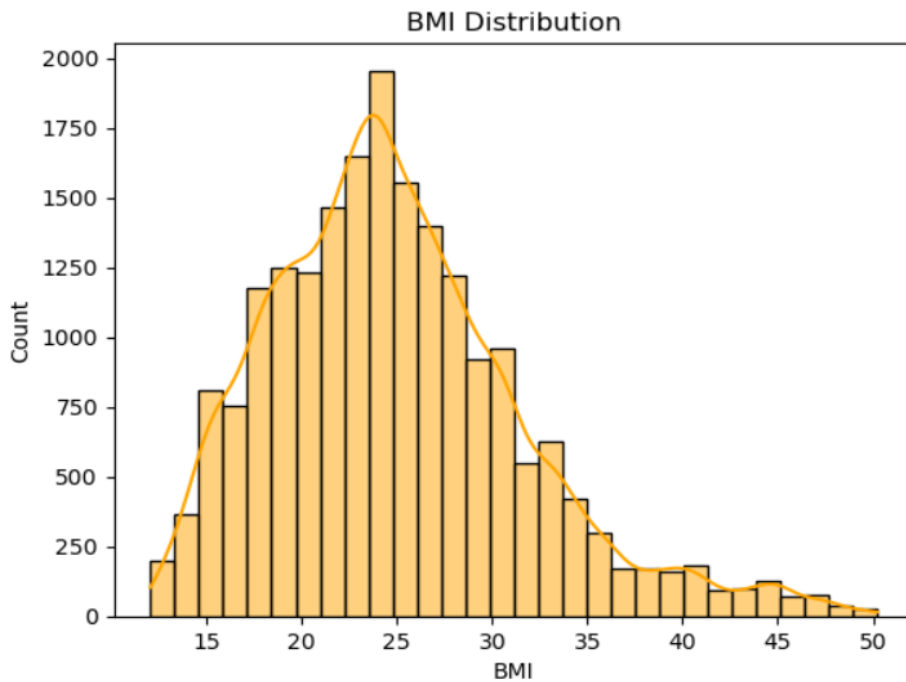
9.Results

1. What is the age distribution of users in the dataset?



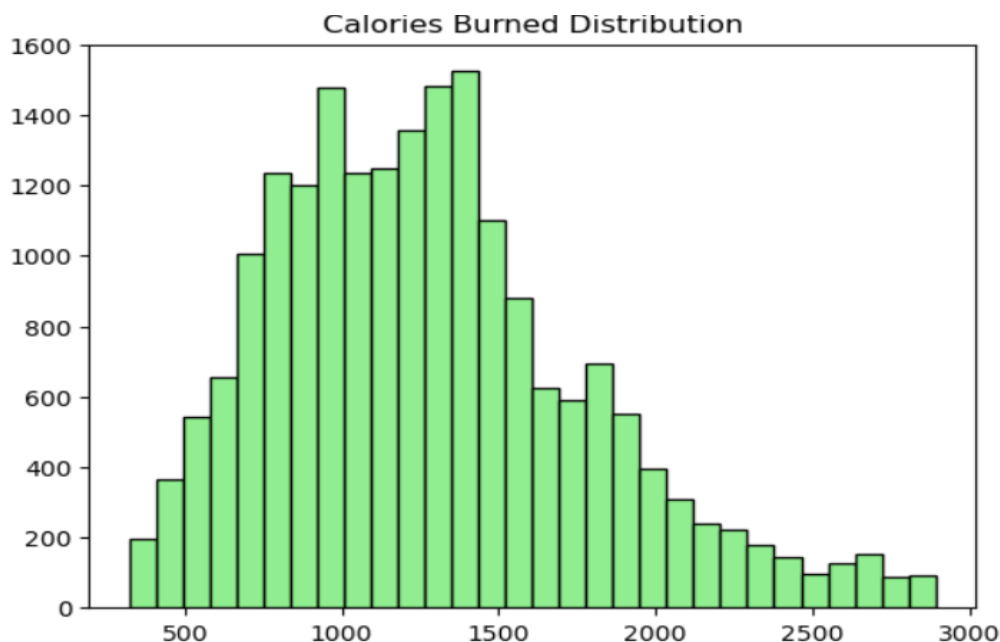
Most users are aged **20–40 years**, showing the dataset mainly includes young adults.

2.What is the BMI range of most users?



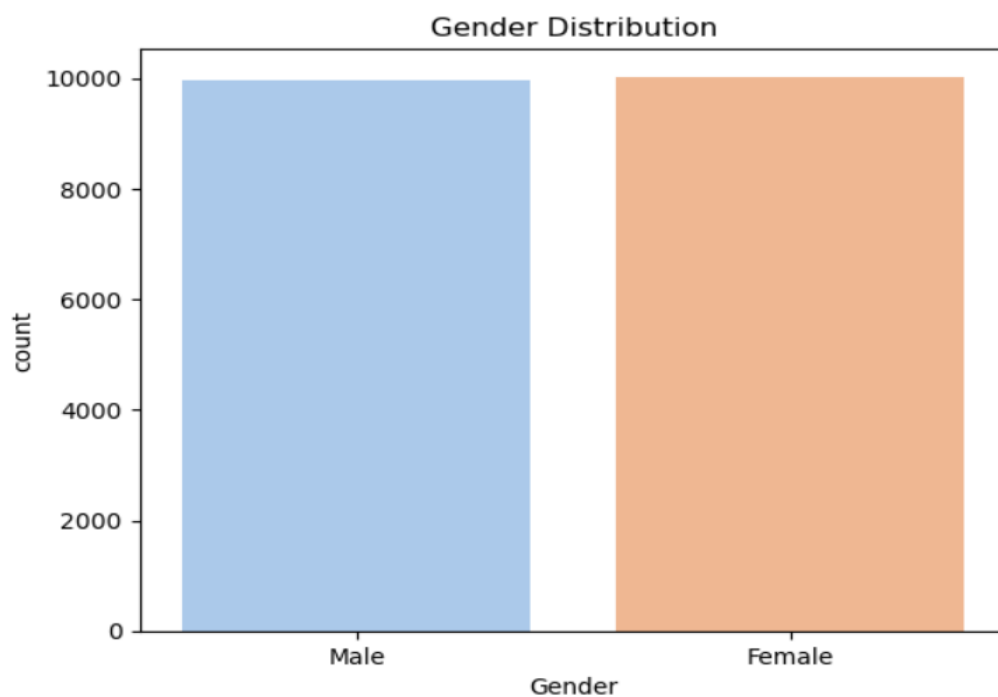
The BMI of most users lies between 18 and 25, indicating healthy weight levels.

3.How many calories do users typically burn in a session?



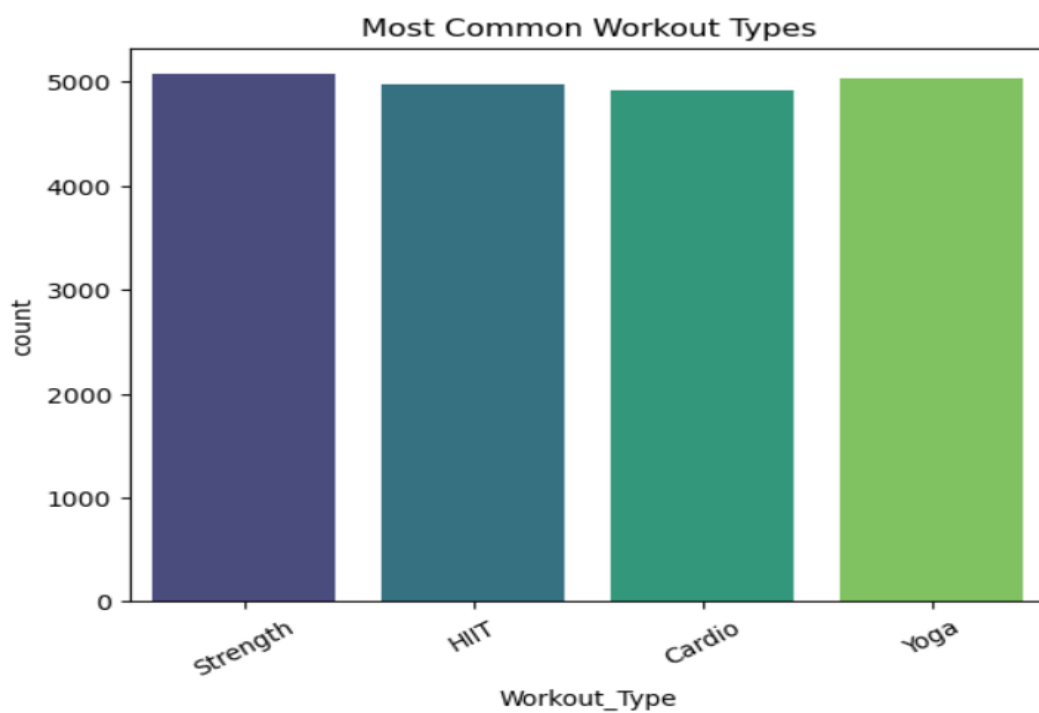
The Calories_Burned column shows most workouts are moderate intensity, burning around 150–400 kcal.

4. What is the gender distribution among users?



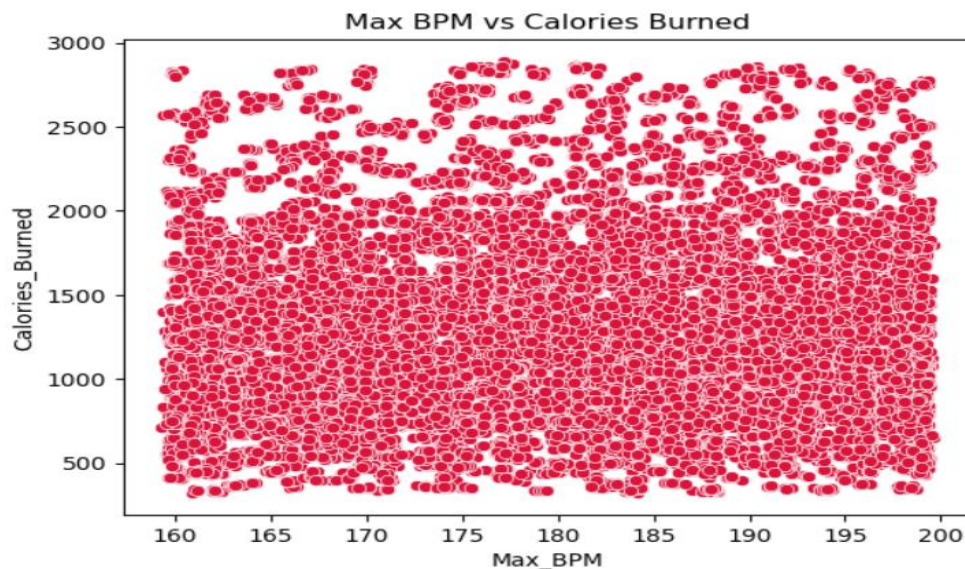
The Gender distribution is almost equal, with a slight majority of females.

5. Which workout types are most commonly performed?



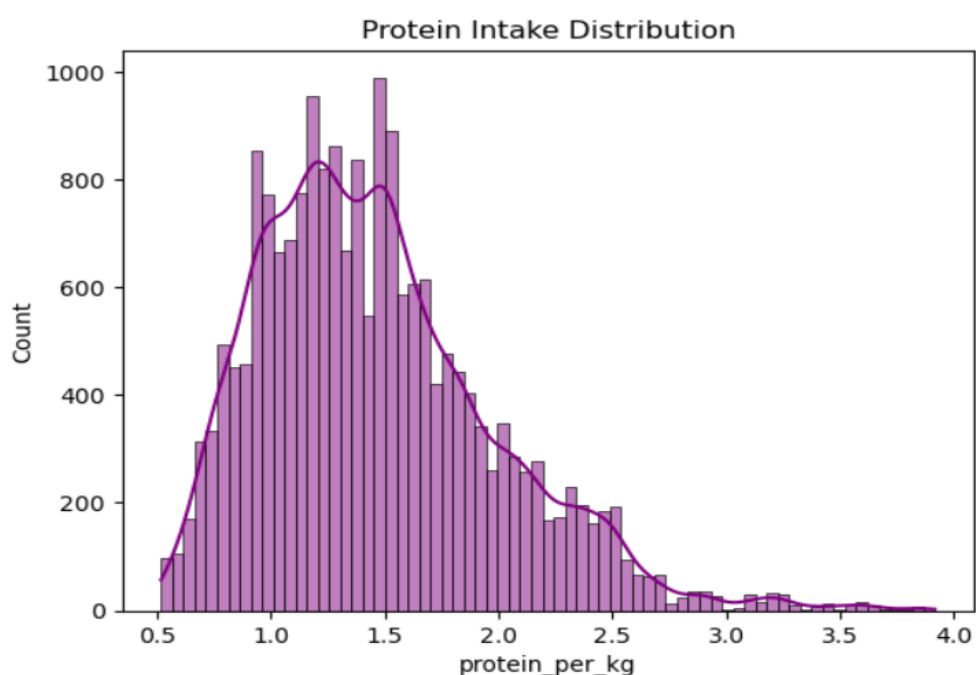
The most common Workout_Type is Cardio, followed by Strength Training.

6. What is the relationship between Max BPM and Calories Burned?



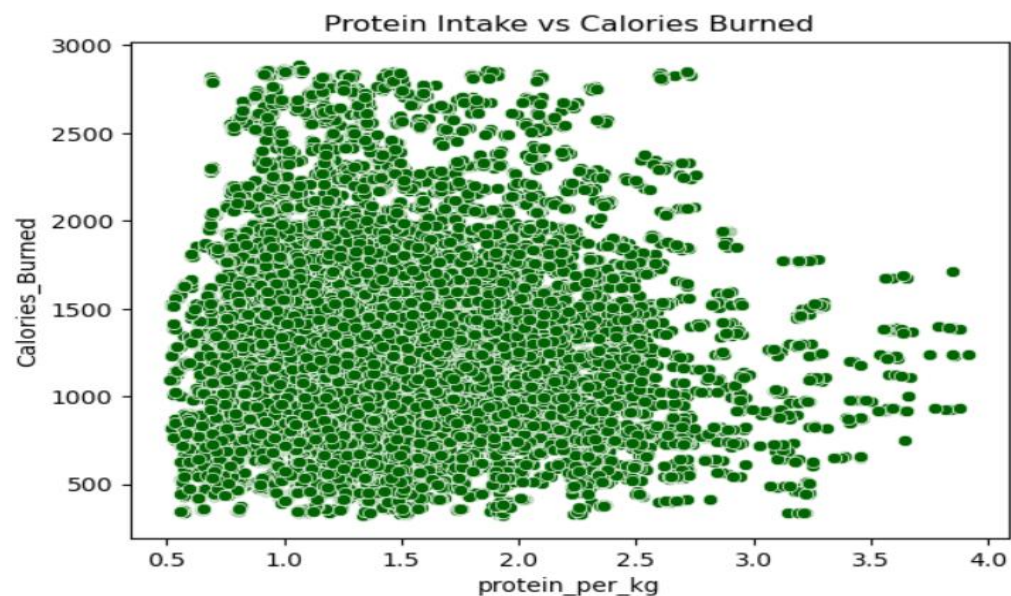
Higher Max BPM leads to more calories burned — intense workouts burn more energy.

7. How is protein intake distributed among users?



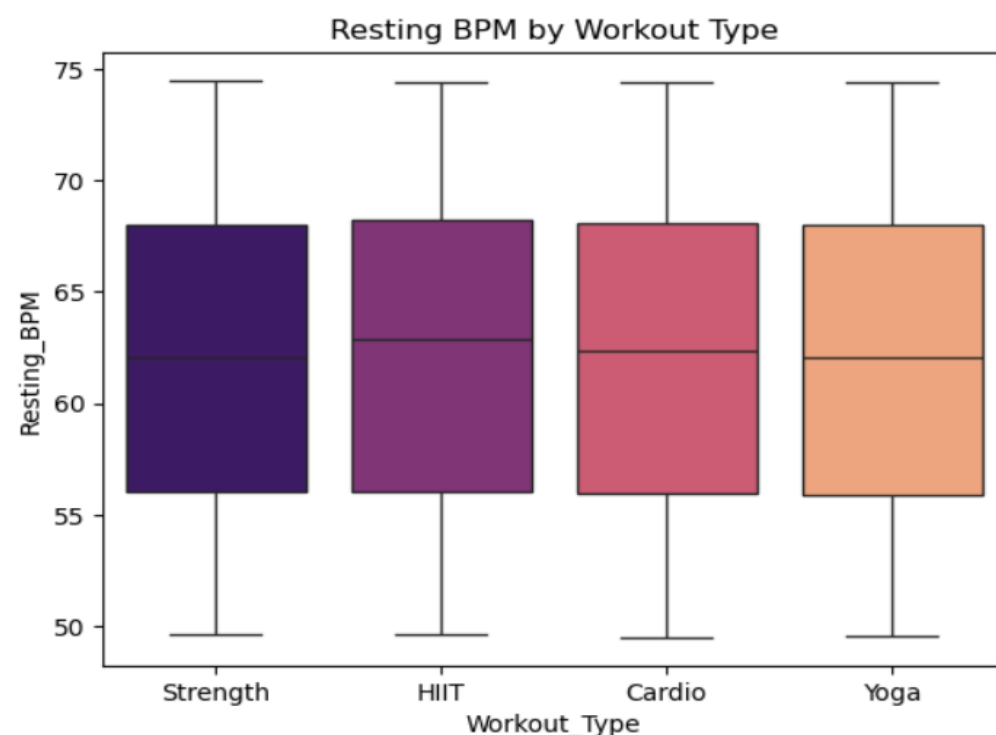
Most users have moderate protein intake, showing balanced diets.

8 Is there any relationship between protein intake and calories burned?



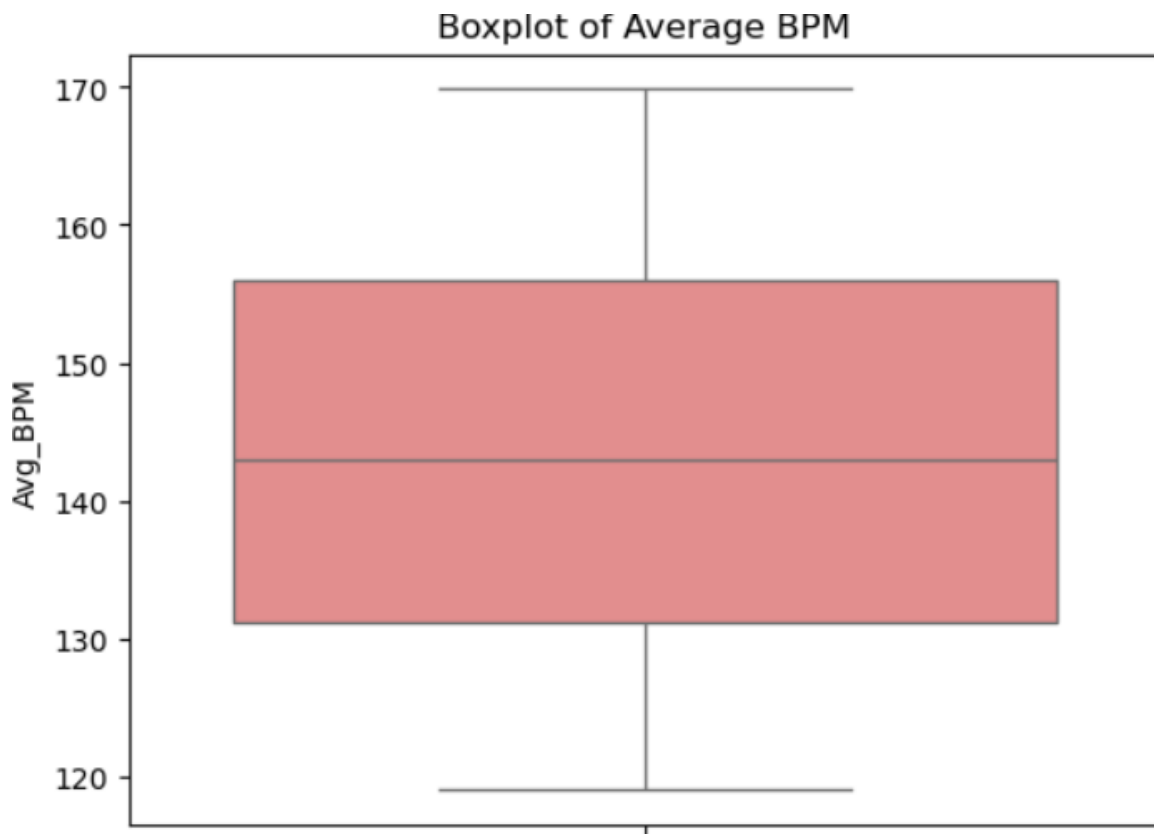
Slight positive link — higher protein intake helps burn more calories.

9. How does Resting BPM differ across workout types?



Cardio/HIIT users have lower Resting BPM, showing better heart fitness.

10 What is the spread of Average BPM among users?



Average BPM is around 100–130, showing moderate workout intensity.

10.Conclusion

This project analyzed fitness and health-related data to understand user behavior, workout patterns, and health indicators. The analysis revealed that most users are young adults with a healthy BMI who regularly engage in moderate-intensity workouts such as cardio and strength training.

The exploratory data analysis showed strong relationships between heart rate (BPM) and calories burned, indicating that higher-intensity exercises result in greater energy expenditure. It was also observed that users with higher protein intake tend to perform better in calorie burning, and those involved in cardio workouts have a lower resting BPM, reflecting better cardiovascular fitness.

Overall, the dataset highlights that maintaining a balanced diet, regular workouts, and monitoring heart rate can lead to improved fitness levels. These insights can help individuals and fitness trainers design better, more personalized fitness plans to achieve optimal health outcomes.